STRATEGIC OPERATIONS BRIEF · ET AI HACKATHON 2026 · PS2

SENTINEL

Signal to executable recommendation in **under 5 minutes.**

A Claude-powered crisis-to-decision engine for India's crude-oil supply chain: it **watches** geopolitical & maritime signals, **simulates** disruption with transparent math, and **decides** — ranked, grade-checked procurement reroutes.

88% import dependence	**43**% via Strait of Hormuz	**9.5**d strategic SPR cover	**<5**min signal → decision

India is one chokepoint away from a price shock.

IMPORT DEPENDENCE

88 %

of crude consumption imported

NET CRUDE IMPORTS

4.71 mb/d

≈ 5.3 demand − 0.59 domestic

VIA HORMUZ

40–45 %

of imports through one strait

STRATEGIC SPR

9.5 days

strategic cover (~65d incl. commercial)

STABILIZATION GAP

~47 days

manual response lag today

REFINING AT RISK

~5 mb/d

nameplate capacity exposed

The cascade — in days, not months

1. **Chokepoint event** → war-risk insurance spikes, transits slow
2. **Refinery run-cuts** as Gulf grades stop arriving
3. **Landed-cost spike** → spot premium + rerouted freight
4. **Retail pass-through** hits the pump in ₹/litre

Today the response is **manual, siloed and slow** — and every day of delay burns scarce reserve cover.

source: ET Hackathon brief PS2 · PPAC ready-reckoner

WHY IT'S HARD TODAY · 03

The signals exist. The decision loop doesn't.

SIGNALS · SILOED

Scattered inputs

News, AIS vessel tracks, prices and sanctions live in separate tools. No single number says "Hormuz is heating up," and no lead-time edge before the price move.

SCENARIOS · OPAQUE

Spreadsheet planning

Impact analysis is buried in spreadsheets — assumptions hidden, un-auditable, and too slow to re-run under crisis time pressure.

PROCUREMENT · TRIBAL

Gut-feel reroutes

Which grade fits which refinery, at what cost, arriving when? It lives in a few experts' heads — not a repeatable, checkable process.

Nobody owns the continuous path from **a signal** to **an executable decision**. That handoff gap **is** the ~47-day stabilization lag. Sentinel closes it into one loop.

Cost of delay: with 9.5 days of strategic cover, decision **time** is the scarce resource — not data, and not compute.

One continuous loop: Watch → Simulate → Decide.

01 · WATCH

Live corridor risk

Every event is classified and folded into a per-corridor risk score (0–100) with a full **evidence trail** — time-decayed, deterministic, auditable.

02 · SIMULATE

Auto-run scenarios

Cross a risk threshold and the mapped disruption scenario fires automatically — computing supply gap, refinery shortfalls, SPR runway and cost, each with its **formula shown**.

03 · DECIDE

Executable reroutes

Ranked (crude, route) cards with volume, ETA, $/bbl delta, **grade-compatible refineries** and caveats — exportable as a one-page procurement brief.

SIGNAL DETECTION
lead time before the price spike

SCENARIO FIDELITY
testable assumptions, unit-tested

EXECUTABILITY
grade-checked alternatives

RESPONSE TIME
<5 min end-to-end

Aligned to the four judging focus areas the brief calls out — lead time, fidelity, executability, response time.

A transparent pipeline — Claude for extraction, deterministic math for decisions.

SIGNAL INGESTION · CACHE-FIRST

GDELT
news · live+cached

AISStream
vessels · live/synth

Yahoo
Brent · live

OFAC
sanctions

▼

AGENT LAYER · CLAUDE / GEMINI / MOCK

1 **Extraction**

2 **Risk**

3 **Scenario**

4 **Procure**

▼

Storage
SQLAlchemy · evidence jsonb

API
FastAPI · REST + WS

Frontend
React · 3 screens

CONFIG · SOURCE OF TRUTH

config/*.yaml — corridors, refineries (API°/sulfur windows), crude assays, and **every coefficient** with a cited source. No magic numbers in code.

STACK

Python 3.11 · FastAPI · SQLAlchemy 2.0 · Pydantic v2 · React + Vite · TypeScript · Tailwind · websocket

Why deterministic? The LLM only **extracts**. Risk, scenario and procurement are transparent, unit-tested math — so every output is reproducible and defensible.

Every score is inspectable.

Tanker traffic on a **real coastline basemap** (Natural Earth), corridor arcs colored by live risk. Click a corridor → the **evidence trail**: which headlines moved the score, with weights and time-decay. A **live GDELT feed**, classified by the extraction agent, streams alongside.

live Brent · live AIS · AISStream · live GDELT news · OFAC sanctions

Real feeds where free & keyless (Brent, GDELT); AISStream vessels with a free key; synthetic clearly labelled in the offline demo.

STRAIT OF HORMUZ · BASELINE

45 / 100

WATCH — structurally tense, no active events

CORRIDOR	SHARE	RISK
Strait of Hormuz	43%	45
Red Sea / Suez	11%	40
Cape of Good Hope	14%	15
Atlantic / US Gulf	9%	10
Pacific / Far East	6%	12

LIVE DEMO · ESCALATION · 07

9 headlines. Hormuz climbs 45 → 76. The scenario fires itself.

45	50	54	55	59	66	66	72	76

"Iran threatens to mine the Strait of Hormuz"

▲ crosses 70 → auto-trigger

```
weighted_sum = Σ severity × type_weight × decay
decay = 0.5 ^ (age_h / 36)
score = baseline + (100-baseline) × logistic01(Σ)
```

HALF-LIFE
36 h

AUTO-TRIGGER
≥ 70

Deterministic & reproducible — no ML, no black box. Each event's weight and decay is stored, so the climb is fully explainable after the fact.

Cascading impact — every number carries its formula.

SUPPLY GAP	SPR RUNWAY VS GAP	LANDED COST Δ	RETAIL PASS-THROUGH
1,013 kb/d	**300** days	**+$30.5** /bbl	**₹5.55** /L
21.5% of net imports	reserve ÷ uncovered gap	Brent $82 → $112.5	est. pump impact

```
gap = Σ net_imports × share × (1-mult) = 1,013
runway = 303,795 / 1,013 = 300 d
Δ$/bbl = premium(24.5) + shock(6) = 30.5
₹/L = 30.5 × 0.52 × 0.35 = 5.55
```

REFINERY (SHORTFALL)	KB/D	%CAP
Jamnagar DTA (RIL)	223	34%
Jamnagar SEZ (RIL)	197	28%
Mangalore (MRPL)	110	37%
Vadinar (Nayara)	102	25%
Panipat, Paradip, ...	381	–

RIPPLE INTO THE WIDER ECONOMY → **+$52 bn/yr** import bill **−0.43 pp** GDP growth **+1.22 pp** current-account **33/100** power-sector stress

Pure functions, unit-tested against hand-computed values. Every coefficient (SPR cover, spot-premium curve, GDP & power sensitivities) lives in **assumptions.yaml** with a source.

Not ML output — a card a procurement head can act on.

1 · Urals — −$0.5/bbl

Volume	255 kb/d
ETA	24 days
Grade-fit	9 refineries

⚠ sanctions / price-cap flag

2 · Tupi — +$7.9/bbl

Volume	186 kb/d
ETA	30 days
Grade-fit	9 refineries

no sanction risk

3 · Bonny Light — +$9.4/bbl

Volume	216 kb/d
ETA	30 days
Grade-fit	9 refineries

light sweet

Ranked on cost + transit + availability + a **deterministic API°/sulfur grade check** per shortfall refinery — then one click exports a markdown **procurement brief**. Signal → recommendation: 4 minutes.

Grade-compat, concretely: Numaligarh (0.5%S max) **rejects** Arab Heavy (2.9%S); Jamnagar (4%S) **accepts** Basrah Heavy (3.9%S). Infeasible cargoes never make the list.

Auditable by construction.

- **Deterministic risk + scenario models** — not ML; every score reproducible and explainable.
- **Pure-function scenario engine** unit-tested against hand-computed expected values.
- **Strict Pydantic schemas** on the LLM step; provider-agnostic (Gemini / Claude / mock).
- **Full evidence trail** persisted per score — which events, which weights, what decay.
- **Runs fully offline** from cache — the demo never touches a live API for its critical path.

BACKEND TEST SUITE

26 passing

flow model · risk · procurement · extraction · AIS parser

LLM ROLE

extract only

MAGIC NUMBERS

zero

EVIDENCE TRAIL (SAMPLE)

`missile_strike · hormuz · sev 0.85 × w 1.6 × decay 1.0 =` **`+1.36`**

Lead time is the buffer.

RESPONSE TIME

days → <5 min

signal → executable reroute vs a ~47-day manual gap

EVERY RECOMMENDATION

grade-checked

API°/sulfur validated per refinery — no infeasible cargoes

With ~9.5 days of strategic cover, the scarce resource in a crude crisis is **decision time**. Compressing signal-to-reroute protects SPR runway and blunts the ₹/L pass-through before it reaches the pump.

DIMENSION	MANUAL TODAY	WITH SENTINEL
Detect disruption	after price move	lead-time edge
Scenario impact	hours, opaque	seconds, sourced
Reroute options	gut-feel	ranked + checked
Auditability	none	full trail

One engine. Any import-dependent chokepoint.

LNG regas terminals	**Coal** seaborne thermal	**Fertilizer** urea / DAP	**Edible oils** palm / soy

The **corridor / commodity / facility** model is commodity-agnostic — swap **config/*.yaml** and the same watch-simulate-decide engine applies. The signal, risk and scenario layers don't change.

PRODUCTION PATH (DROP-IN)

`Postgres + pgvector` `LangGraph` `live AIS + GDELT` `SPR optimiser`

From prototype to standing capability.

NEXT
SPR drawdown optimiser · full supply-chain **digital twin**

THEN
Backtest module (replay historic standoffs to prove lead time) · live AIS + GDELT at production scale

88%
import dependence

9.5d
strategic cover

26
tests passing

<5min
signal → decision

The ask: **a pilot with a refiner or PPAC.**

SENTINEL